

Question 1 (12 marks)

Marks

(a) Evaluate $\frac{12.9}{\sqrt{6.7 \times 3.4}}$ correct to 3 significant figures.

2

(b) Factorise $1 - 8y^3$.

2

(c) Find the value of $\frac{\log_3 8}{\log_3 2}$.

2

(d) Find a primitive of $5 + \sin 2x$

2

(e) Find the values of x for which $x^2 - 6x + 5 > 0$.

2

(f) Solve the simultaneous equations :

2

$$2x + y = 3$$

$$x - 2y = 4$$

Question 2 (12 marks) Start question on a new page.

Marks

(a) Differentiate with respect to x :

(i) $(x + 1)^7$

1

(ii) $x \tan x$

2

(iii) $\log_e \left(\frac{x}{x-1} \right)$

2

(b) Find:

(i) $\int \frac{x}{x^2 + 6} dx$

2

(ii) $\int \frac{3}{e^{2x}} dx$

2

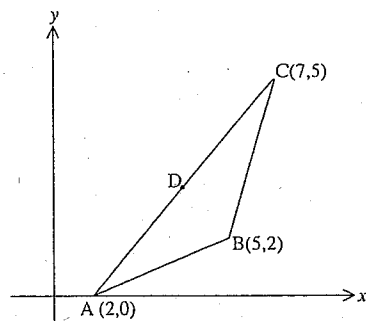
(c) Evaluate $\int_1^e \left(\frac{2}{x} + \frac{x}{2} \right) dx$ leaving your answer in exact form.

3

Question 3 (12 marks) Start question on a new page.

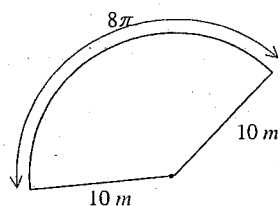
Marks

- (a) The points $A(2,0)$, $B(5,2)$ and $C(7,5)$ are joined to form a triangle as shown below. D is the midpoint of AC .



- (i) Find the length of AC 1
- (ii) Find the co-ordinates of D 1
- (iii) Find the slope of DB , and prove that it is perpendicular to AC 2
- (iv) BD is extended to E , so that $BD = DE$. Find the co-ordinates of the point E . 1
- (v) Find the area of the quadrilateral $ABCE$ 2

(b)



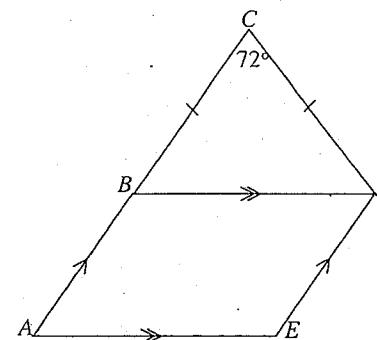
The diagram shows a garden bed in the shape of a sector. The arc length is 8π metres and the radius is 10 metres

- (i) Show that the angle of the sector is $\frac{4\pi}{5}$ radians 1
- (ii) Calculate the area of this garden bed. 2
- (iii) The garden bed is to be planted with red and yellow tulips. If the tulips can be planted at 15 per square metre, how many tulips can be planted? 1
- (iv) Assuming all tulips flower, what is the expected number of red tulips if the probability of producing a red flower is 0.6? 1

Question 4 (12 marks) Start question on a new page.

Marks

(a)



A , B and C are collinear points.
 $BD \parallel AE$, $AB \parallel ED$, $BC = BD$
 and $\angle BCD = 72^\circ$

Copy this diagram on your answer sheet.

Find the size of $\angle DEA$, giving reasons.

- (b) Use Simpson's rule with three function values (i.e. one application) 3

to estimate $\int_1^5 \log_e x \, dx$.

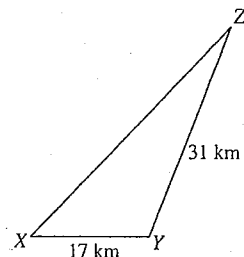
- (c) Solve $4^x - 18(2^x) + 32 = 0$ 3

- (d) Solve $2\cos 2x + \sqrt{3} = 0$ for $0 \leq x \leq 2\pi$ 3

Question 5 (12 marks) Start question on a new page.

Marks

- (a) In the diagram X , Y and Z represent the locations of three towns. The town Y is due east of X and the bearing of Z from Y is 046° .



- (i) Find the size of $\angle XYZ$.
 (ii) Find the distance XZ to 1 decimal place.
 (iii) What is the bearing of Y from Z ?

- (b) The roots of the equation $x + \frac{1}{x} = 7$ are α and β . Find the value of:

- (i) $\alpha + \frac{1}{\alpha}$
 (ii) $\alpha + \beta$

- (c) (i) Show that $\frac{3x+4}{x+1} = \frac{1}{x+1} + 3$

Hence:

- (ii) Sketch the graph of $y = \frac{3x+4}{x+1}$ showing all the important features. (Do not find stationary points).

- (iii) Find the exact area of the region bounded by the curve $y = \frac{3x+4}{x+1}$, the x and y axes, and the line $x = 2$.

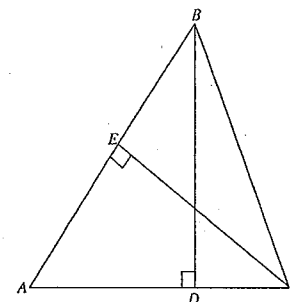
Question 6 (12 marks) Start question on a new page.

Marks

- (a) Given that $\log_a 3 = 0.68$ and $\log_a 2 = 0.42$, find $\log_a 18$

- (b) Find the limiting sum of the series $\frac{9}{8} + \frac{3}{4} + \frac{1}{2} + \dots$

- (c)



The diagram shows $BD \perp AC$ and $CE \perp AB$

- (i) Copy this diagram into your answer booklet and prove $\triangle ECA \sim \triangle DBA$

- (ii) If $AB = 10$ cm, $BD = 7$ cm and $AC = 6$ cm find the length of CE .

- (d) The rate of water flowing, R litres per hour, into a pond is given by

$$R = 65 + 4t^{\frac{1}{3}}$$

- (i) Calculate the initial flow rate.

- (ii) If initially there was 15 litres in the pond, find the volume of the water in the pond when 8 hours have elapsed.

Question 7 (12 marks) Start question on a new page.

Marks

(a) Evaluate $\sum_{k=3}^5 2^{4-k}$

1

(b) A particle moves in a straight so that its distance x , in metres, from a fixed point O at time t , in seconds, is given by

$$x = 5t + \log_e(1 - 2t), \quad 0 \leq t \leq \frac{1}{2}.$$

(i) Find the initial velocity and acceleration of the particle.

4

(ii) When does the particle come to rest?

2

(c) A parabola has the equation $x^2 = -12y$

(i) Find the co-ordinates of the vertex of the parabola

1

(ii) Write down the focus of the parabola

1

(iii) Find the equation of the tangent of the parabola at the point where $x = 6$.

2

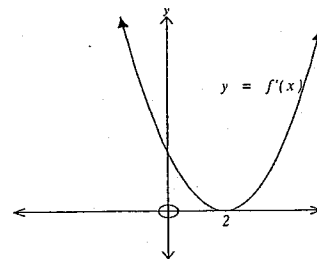
(iv) Find the co-ordinates of Y , the point where the tangent cuts the y -axis

1

Question 8 (12 marks) Start question on a new page.

Marks

(a) The figure shows the graph of $y = f'(x)$



The curve $y = f(x)$ has a stationary point at $(2, 0)$.
What is the nature of this stationary point?

1

(b) Consider the curve $y = \frac{1}{x} e^{-x}$:

(i) For what values of x is the function defined?

1

(ii) Describe the behaviour of the function as x :

2

(α) approaches zero

(β) increases indefinitely

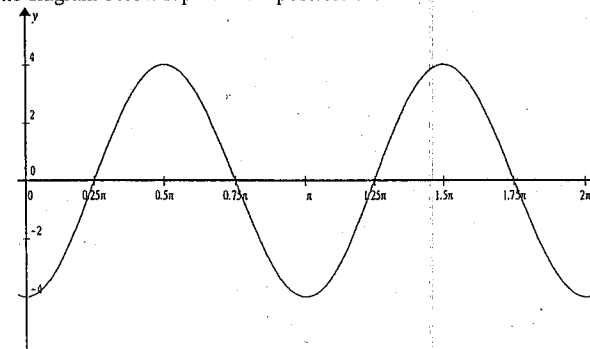
(iii) Find any stationary points and determine their nature.

3

(iv) Sketch the curve of this function

2

(c) The diagram below represents a possible sine or cosine curve.



(i) Give the amplitude

1

(ii) Give the period

1

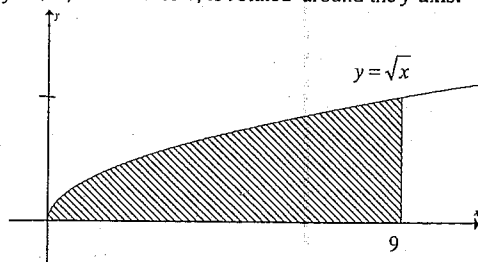
(iii) Write down the possible equation of the curve

1

Question 9 (12 marks) Start question on a new page.

Marks

- (a) Find the volume of the solid formed when the shaded area under the curve $y = \sqrt{x}$, shown below, is rotated around the y -axis.



4

- (b) (i) Sketch the curve $y = 3\sin \frac{\pi x}{2}$ for $-2 \leq x \leq 4$.

1

- (ii) Draw on your diagram a line, clearly labelled, which can be used to solve the following equation:

2

$$\sin \frac{\pi x}{2} - \frac{x}{3} = 0$$

- (iii) Determine the number of solutions to the equation

1

$$\sin \frac{\pi x}{2} - \frac{x}{3} = 0 \text{ over the domain } -2 \leq x \leq 4.$$

- (c) In a game of chess between two players X and Y , both of approximately equal ability, the player with the *White* pieces, having the first move, has a probability of 0.5 of winning, and the probability that the player with the black pieces for that game winning is 0.3.

- (i) What is the probability that the game ends in a draw?

1

- (ii) The two players X and Y play each other in chess competition, each player having the *White* pieces once. In the competition the player who wins the game scores 3 points, the player who loses the game scores 1 point and in a draw each player receives 2 points.

By drawing a tree diagram, or otherwise, find the probability that, as a result of these two games,

- (α) X scores 6 points.

1

- (β) X scores less than 4 points.

2

Question 10 (12 marks) Start question on a new page.

Marks

- (a) The number N of a certain species is falling according to $N = N_0 e^{-0.03t}$ where t is in days and N_0 is the initial number of species present.

- (i) Show that $N = N_0 e^{-0.03t}$ is a solution to the differential equation

1

$$\frac{dN}{dt} = -0.03N.$$

- (ii) How long, to the nearest day, will it take for the number of species to halve?

1

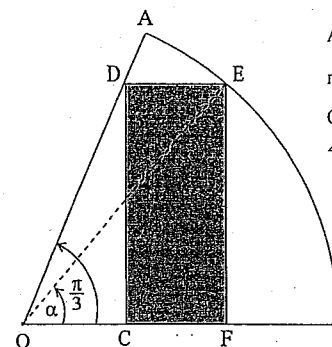
- (iii) Find, in terms of N_0 , the rate of change at the time when the number of species has halved.

1

- (iv) Find the number of days, to the nearest whole number, for the number of species to fall to just below 5% of the initial number.

2

- (b)



AOB is a sector of a circle with centre at O and radius r such that $\angle AOB = \frac{\pi}{3}$.

CDEF is a rectangle drawn in the sector and $\angle EOF = \alpha$ as shown in the diagram.

NOT TO SCALE

- (i) Show that $CF = r \cos \alpha - \frac{r \sin \alpha}{\sqrt{3}}$

2

- (ii) Given that $\frac{1}{2} \sin 2\alpha = \sin \alpha \cos \alpha$, show that the area of rectangle CDEF

$$\text{can be expressed as } A = r^2 \left(\frac{1}{2} \sin 2\alpha - \frac{\sqrt{3}}{3} \sin^2 \alpha \right)$$

2

- (iii) Find the value for α which will produce the rectangle of maximum area.

3

QUESTION 1

Solution of 2006 24mt Trial NSBHS

a) $2.70279 \div 2.70$ (3 sign. figures) ✓✓

b) $1 - 8y^3 = (1 - 2y)(1 + 2y + 4y^2)$ ✓✓

c) $\frac{\log_3 8}{\log_3 2} = \frac{3 \log_3 2}{\log_3 2}$ ✓
 $= 3$ ✓

d) $\int (\sin 2x + 5) dx = -\frac{1}{2} \cos 2x + 5x + C$ ✓✓

e) $(x^2 - 6x + 5) > 0$ ✓
 $(x - 5)(x - 1) > 0$ ✓
 $x < 1$ or $x > 5$

f) $2x + y = 3$
 $x - 2y = 4$
 $x = 4 + 2y$
 $2(4 + 2y) + y = 3$
 $8 + 4y + y = 3$
 $5y = -5$
 $y = -1$
 $x = 2$ ✓✓

QUESTION 3

i) A(2,0) C(7,5)
 $AC = \sqrt{(7-2)^2 + (5-0)^2}$
 $= \sqrt{50} = 5\sqrt{2}$ ✓

ii) D($\frac{9}{2}, \frac{5}{2}$) ✓

iii) $m_{DB} = \frac{\frac{5}{2} - 0}{\frac{9}{2} - 2}$
 $= -1$ ✓

$m_{AC} = \frac{5-0}{7-2}$
 $= 1$

$m_{DB} \times m_{AC} = -1$ ✓
 $\therefore DB \perp AC$

iv) E(4,3) ✓

ABCE is a kite (diag. bisect at 90°)

Area = $\frac{1}{2} \times AC \times EB$

$EB = \sqrt{(4-5)^2 + (3-2)^2}$
 $= \sqrt{2}$

Area = $\frac{1}{2} \times 5\sqrt{2} \times \sqrt{2}$
 $= 5$ sq units ✓

i) $l = r\theta$
 $8\pi = 10\theta$
 $\therefore \theta = \frac{4\pi}{5}$ ✓

ii) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 10^2 \times \frac{4\pi}{5}$
 $= 40\pi$ ✓

iii) No. of tulips = $40\pi \times 15 = 1884$ ✓

iv) No. of red = 0.6×1884
 $= 1130$ ✓

QUESTION 2

a) i) $y = (x+1)^7$
 $\frac{dy}{dx} = 7(x+1)^6$ ✓

ii) $y = x \tan x$
 $\frac{dy}{dx} = 1 \cdot \tan x + x \sec^2 x$ ✓✓

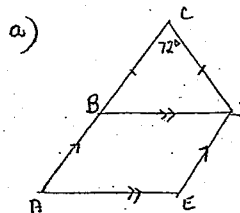
iii) $y = \log_e \left(\frac{x}{x-1} \right)$
 $= \log_e x - \log_e (x-1)$
 $\frac{dy}{dx} = \frac{1}{x} - \frac{1}{x-1}$ ✓

b) i) $\int \frac{x}{x^2+6} dx = \frac{1}{2} \ln(x^2+6) + C$ ✓✓

ii) $\int 3e^{-2x} dx = -\frac{3}{2} e^{-2x} + C$ ✓✓

c) $\int_1^e \left(\frac{2}{x} + \frac{x}{2} \right) dx = \left[2 \ln x + \frac{x^2}{4} \right]_1^e$
 $= 2 \ln e + \frac{e^2}{4} - \left(0 + \frac{1}{4} \right)$
 $= \frac{7}{4} + \frac{e^2}{4}$ ✓✓

QUESTION 4



$\angle CBD = \frac{180^\circ - 72^\circ}{2}$ (isos. Δ , base $\angle$ s are $\angle$ s)
 $= 54^\circ$ ✓

$\angle ABD = 180^\circ - 54^\circ$ (st. line)
 $= 126^\circ$ ✓

$\therefore \angle AED = 126^\circ$ (opp. $\angle$ s of parallelogram) ✓

b)

x	1	3	5
y	0	1.099	1.609

 $y = \log_e x$ ✓

$\int_1^5 \log_e x dx = \frac{2}{3} [0 + 4 \times 1.099 + 1.609]$ ✓
 $\div 4$ ✓

c) Let $u = 2^x$
 $u^2 - 18u + 32 = 0$
 $(u-16)(u-2) = 0$
 $\therefore u = 16$ or $u = 2$
 $\therefore 2^x = 16$ or $2^x = 2$
 $\therefore x = 4$ or $x = 1$ ✓✓

d) $2 \cos 2x + \sqrt{3} = 0$
 $\cos 2x = -\frac{\sqrt{3}}{2}$
 $2x = \pi + \frac{\pi}{6}, \pi + \frac{\pi}{6}, 3\pi - \frac{\pi}{6}, 3\pi - \frac{\pi}{6}$
 $x = \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{17\pi}{12}, \frac{19\pi}{12}$ ✓✓

QUESTION 5

i) $\angle XYZ = 46^\circ + 90^\circ = 136^\circ$

ii) $XZ^2 = 17^2 + 21^2 - 2 \times 17 \times 21 \times \cos 136^\circ$

$XZ \approx 44.8$

iii) $180^\circ + 46^\circ = 226^\circ$

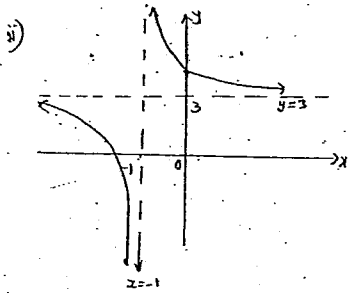
$x + \frac{1}{x} = 7$

i) $x + \frac{1}{x} = 7$

ii) $x^2 + 1 = 7x$
 $x^2 - 7x + 1 = 0$

$\alpha + \beta = -\frac{b}{a}$
 $= 7$

i) $\frac{1}{x+1} + 3 = \frac{1+3(x+1)}{x+1}$
 $= \frac{3x+4}{x+1}$



✓ for asymptotes graph.

ii) $A = \int \frac{1}{x+1} + 3 dx$
 $= [\ln(x+1) + 3x]^2_0$
 $= \ln 3 + 6 - (0 + 0)$
 $= \ln 3 + 6$ so units

QUESTION 7

i) $\sum_{k=3}^5 2^{4-k} = 2 + 2 + 2 = 3\frac{1}{2}$

b) i) $x = 5t + \log_e(1-2t)$

$\frac{dx}{dt} = 5 + \frac{-2}{1-2t}$
 $= 5 - 2(1-2t)^{-1}$ — (1)

$\frac{d^2x}{dt^2} = 2(1-2t)^{-2}$
 $= \frac{4}{(1-2t)^2}$ — (2)

$t=0, v = 3 \text{ m/s}$
 $a = -4 \text{ m/s}^2$

ii) $v=0$
 $5 - \frac{2}{1-2t} = 0$

$5(1-2t) - 2 = 0$
 $3 - 10t = 0$
 $t = \frac{3}{10} \text{ sec}$

c) i) $v(0,0)$

ii) $-12 = 4a$
 $a = -3$
 $\therefore F(0, -3)$

iii) $y = -\frac{x^2}{12}$
 $\frac{dy}{dx} = -\frac{x}{6}$
When $x=6, m_T = -1$
 $y = -3$
 $y - y_1 = m(x - x_1)$

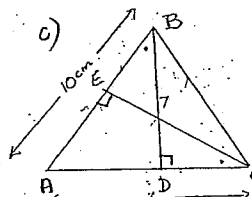
iv) $x=0$
 $y=3$
 $A(0,3)$

QUESTION 6

a) $\log_a 18 = \log_a(2 \times 3^2)$
 $= \log_a 2 + 2 \log_a 3$
 $= 0.42 + 2 \times 0.68$
 $= 1.78$

b) $a = \frac{9}{8}, r = \frac{3}{4} \div \frac{9}{8}$
 $= \frac{2}{3}$

$S = \frac{a}{1-r}$
 $= \frac{\frac{9}{8}}{1-\frac{2}{3}}$
 $= \frac{27}{8}$



i) $\angle A$ is common
 $\angle AEC = \angle BDA = 90^\circ$ (given)
 $\therefore \triangle AEC \parallel \triangle BDA$ equiangular

ii) $\frac{CE}{BD} = \frac{AC}{AB}$ (corres sides of similar Δ s)
 $\frac{CE}{7} = \frac{6}{10}$
 $CE = \frac{6}{10} \times 7 = 4.2$

d) $R = 65 + 4t^{4/3}$
i) $t=0, R = 65 \text{ L/hr}$
ii) $V = \int 65 + 4t^{4/3} dt$
 $= 65t + 4 \times \frac{3}{7} t^{7/3} + C$
 $= 65t + 3t^{7/3} + C$
 $t=0, V=15$
 $\therefore C=15$
 $\therefore V = 65t + 3t^{7/3} + 15$
When $t=8$
 $V = 65 \times 8 + 3 \times 8^{7/3} + 15$
 $= 583 \text{ litres}$

QUESTIONS

a) Horizontal point of inflexion

b) i) all real $x, x \neq 0$

ii) a) $x \rightarrow 0, y \rightarrow \infty$
b) $x \rightarrow \infty, y \rightarrow 0$

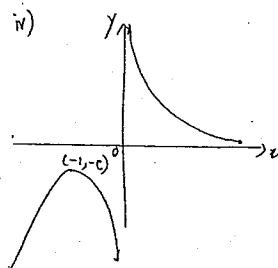
iii) $y = \frac{e^{-x}}{x}$
 $\frac{dy}{dx} = \frac{-e^{-x}x - e^{-x}}{x^2}$

$\frac{dy}{dx} = 0, -e^{-x}x - e^{-x} = 0$
 $-e^{-x}(x+1) = 0$
 $\therefore x = -1$

$y = \frac{e}{-1} = -e$

x	-1	-1	-1
y	$+0$	$-$	$-$

$\therefore \text{Max}(-1, -e)$



c) i) 4
ii) π
iii) $y = -4 \cos 2x$

